

Engineering Reboot 2026

Week-02 Day-06 Engineering Lab Manual v4.0 - Part 3 Documentation, Git Workflow, and Audits

Section 16 - Notes.md

Purpose: Capture technical learning and observations.

Location:

Week-02\Notes\Day-06\Notes.md

Required Sections:

- Concepts Learned
- Key Commands
- Key Observations
- Lessons Learned
- Questions For Future Study

Example entries should reflect Day 06 conditional logic concepts.

Section 17 - Journal.md

Purpose: Daily engineering retrospective.

Location:

Week-02\Journal\Day-06\Journal.md

Required Sections:

- Summary
- Challenges Encountered
- Solutions Applied
- Confidence Assessment
- Reflection
- Tomorrow Preparation

Journal should capture lessons learned during implementation.

Section 18 - Validation Matrix

High Temperature Alert:

85 -> Air conditioning recommended

Temperature Status:

65 -> Temperature acceptable

90 -> Air conditioning recommended

Age Classification:

16 -> Minor

21 -> Adult

Grade Evaluation:

83 -> Grade B

Decision Program:

100 -> Voltage Low

115 -> Voltage Normal

125 -> Voltage High

Section 19 - Evidence Matrix

Required Evidence:

Git-Status.png

high_temperature_alert-Code.png

high_temperature_alert-ExpectedResult.png

temperature_status-Code.png

temperature_status-ExpectedResult.png

age_classification-Code.png

age_classification-Minor.png

age_classification-Adult.png

grade_evaluation-Code.png

grade_evaluation-ExpectedResult.png

decision_program-Code.png

decision_program-Voltage-Low.png

decision_program-Voltage-Normal.png

decision_program-Voltage-High.png

Git-Commit.png

Git-Push-Success.png

GitHub-Remote-Verification.png

Storage:

Week-02\Day-06\Evidence\

Section 20 - Git Workflow Walkthrough

Step 1:

git status

Step 2:

git add .

Step 3:

git commit -m "Day 06 conditionals completed"

Step 4:

git push

Step 5:

Verify GitHub repository synchronization.

Capture:

Git-Commit.png

Git-Push-Success.png

GitHub-Remote-Verification.png

Section 21 - Expected Repository End-State

Week-02

■■■ Exercises

■ ■■■ Day-06

■ ■■■ high_temperature_alert.py

■ ■■■ temperature_status.py

■ ■■■ age_classification.py

■ ■■■ grade_evaluation.py

■ ■■■ decision_program.py

■■■ Notes

■ ■■■ Day-06

■ ■■■ Notes.md

■■■ Journal

■ ■■■ Day-06

■ ■■■ Journal.md

■■■ Day-06

■■■ Evidence

Section 22 - Day 06 Completion Audit

Deliverable Audit:

Verify all five Python files exist.

Documentation Audit:

Verify Notes.md and Journal.md exist.

Evidence Audit:

Verify all screenshots are present.

Git Audit:

Verify commit, push, and GitHub synchronization completed.

Section 23 - Completion Checklist

[] Exercise 01 complete

[] Exercise 02 complete

[] Exercise 03 complete

[] Exercise 04 complete

[] Exercise 05 complete

[] Validation complete

[] Notes complete

[] Journal complete

[] Evidence collected

[] Git commit completed

[] Git push completed

[] GitHub verification completed

Section 24 - Day 07 Preview

Next Topic: Boolean Logic

Concepts:

- AND
- OR
- NOT

Objective:

Combine multiple conditions to create more advanced decision-making programs.